PennState | One Health Microbiome Center

KickStart Workshop

Fall 2026 KickStart Workshop for the analysis of microbial communities

Day 2: Amplicon Data Analysis with Qiime2

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Day 2

- About amplicon sequencing
- Data analysis with Qiime
- Lunch break
- Cont. data analysis with Qiime

Learning Goals

Concepts What is an amplicon, why is a good barcode, and how sequencing it reshaped our view of microbial diversity.

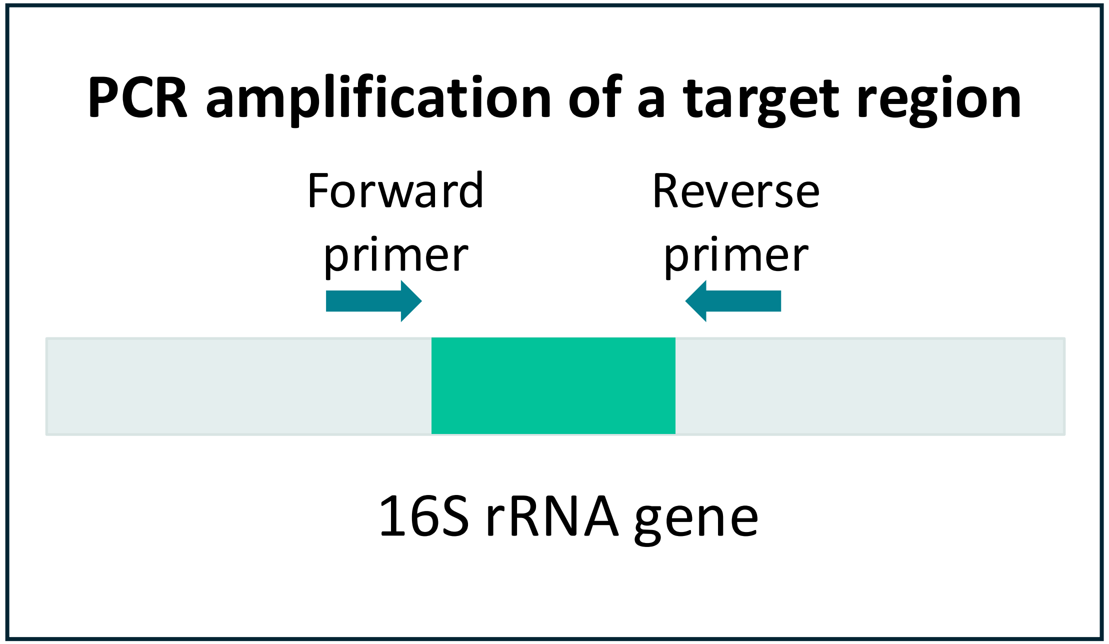
Tools Which software options exist for amplicon analysis, and how to work with QIIME 2.

Data Data and databases. NCBI SRA.

Workflow The QIIME 2 workflow: from raw reads to diversity, taxonomy, and differential-abundance results.

What is an amplicon?

- A DNA fragment produced by PCR **amplification** of one specific, **targeted** region of a genome.
- In microbiome work, that target is a taxonomically informative marker gene.
- Millions of copies of that one region, from every organism in the sample, are amplified and sequenced together.

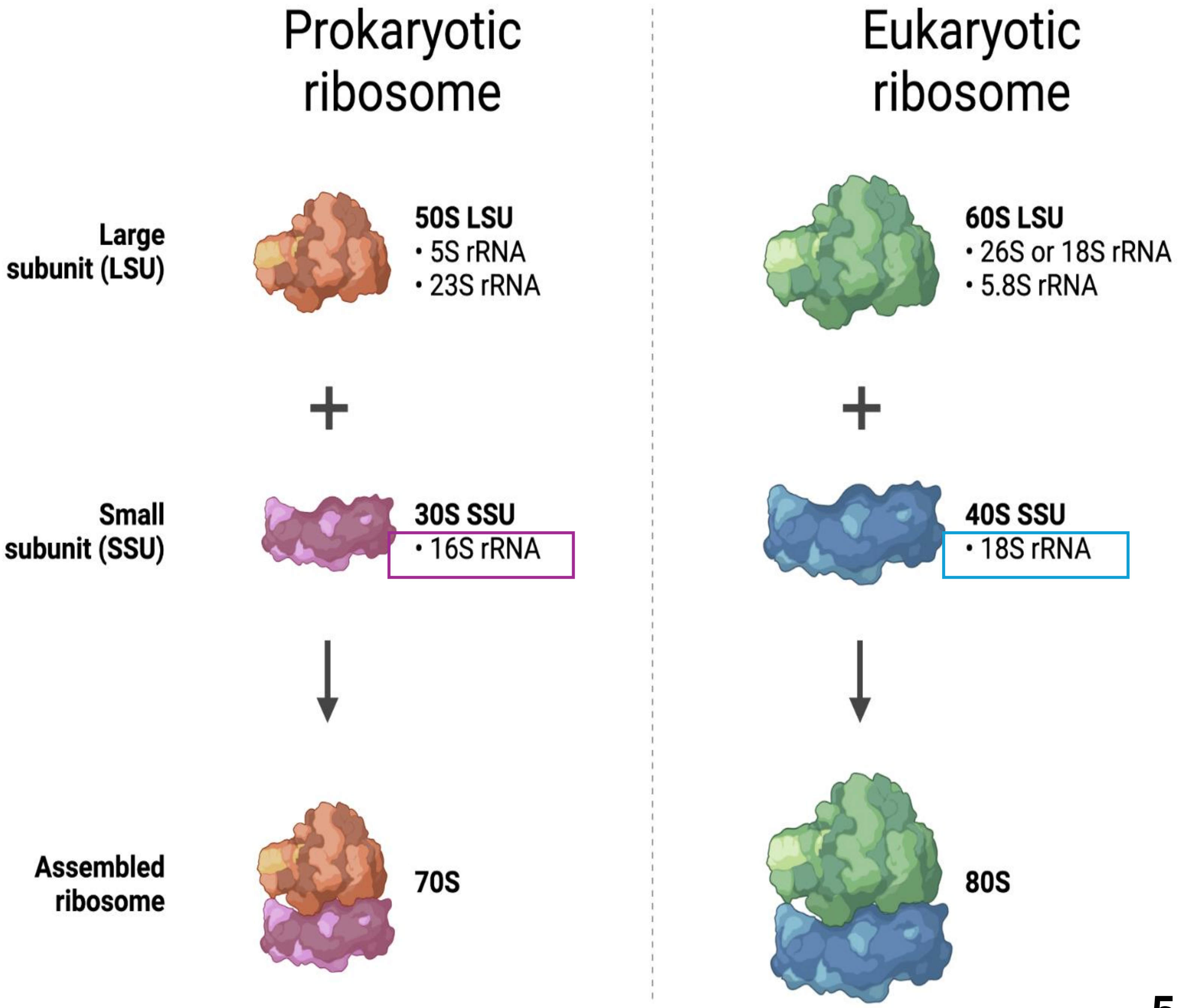


Taxonomically informative marker:

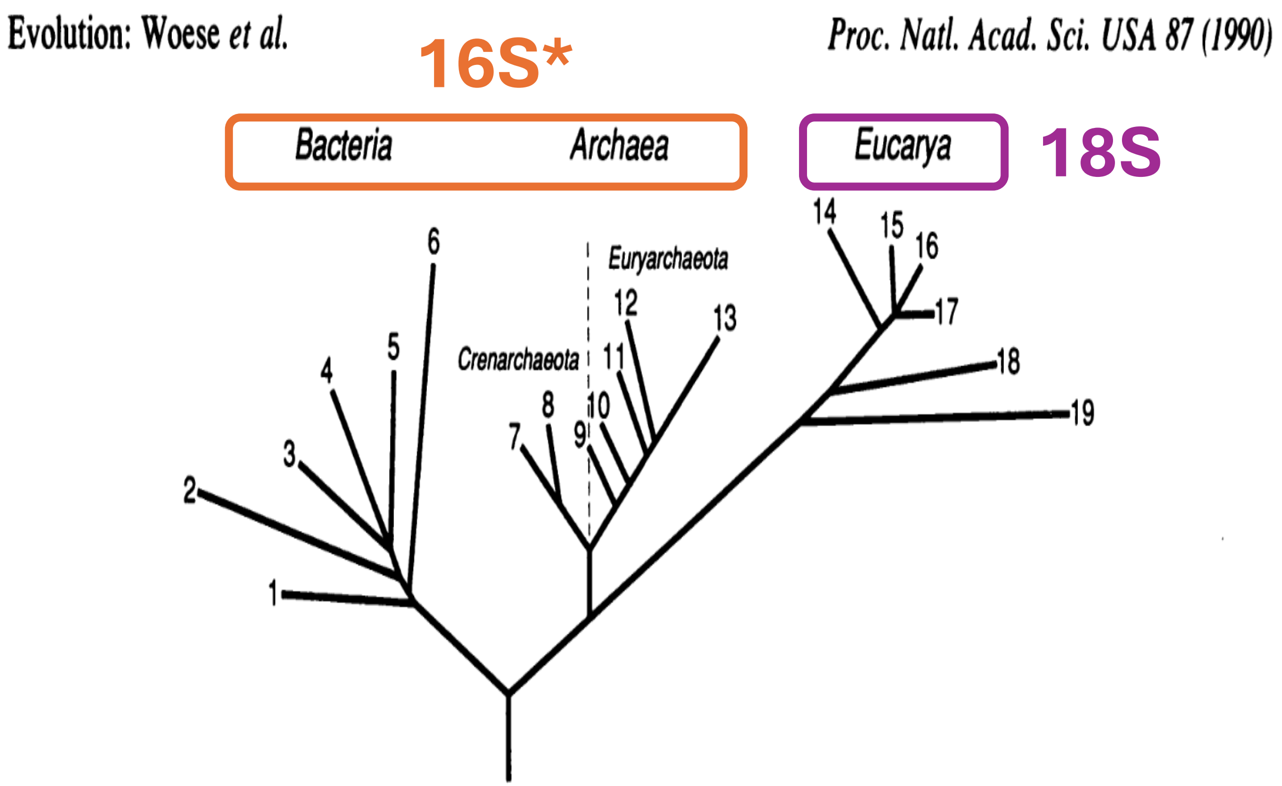
The most common markers are Ribosomal RNA (rRNA)

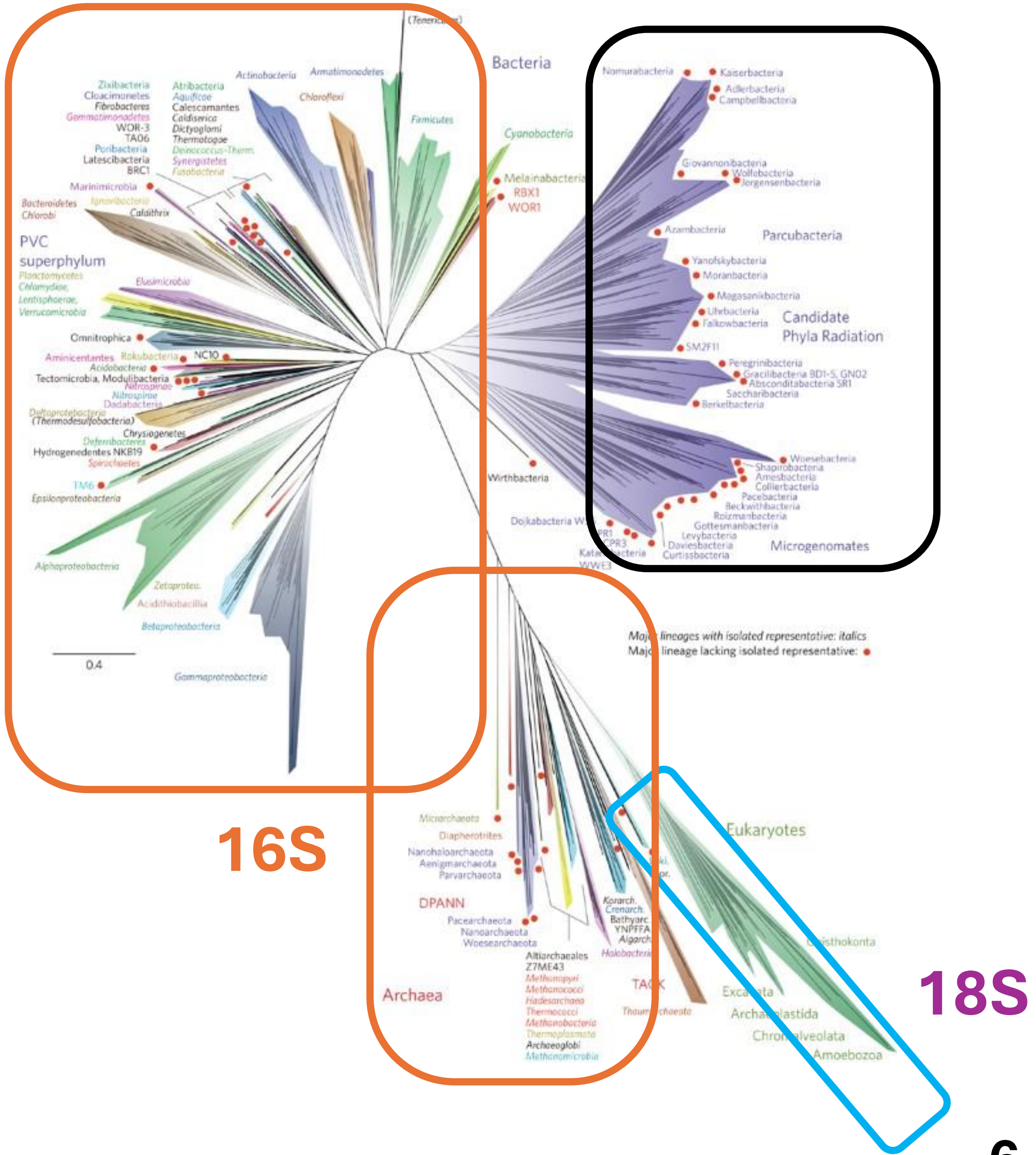
What is the S?
Svedberg unit.

A measure of how fast a particle sediments in a centrifuge.



1990 vs 2016

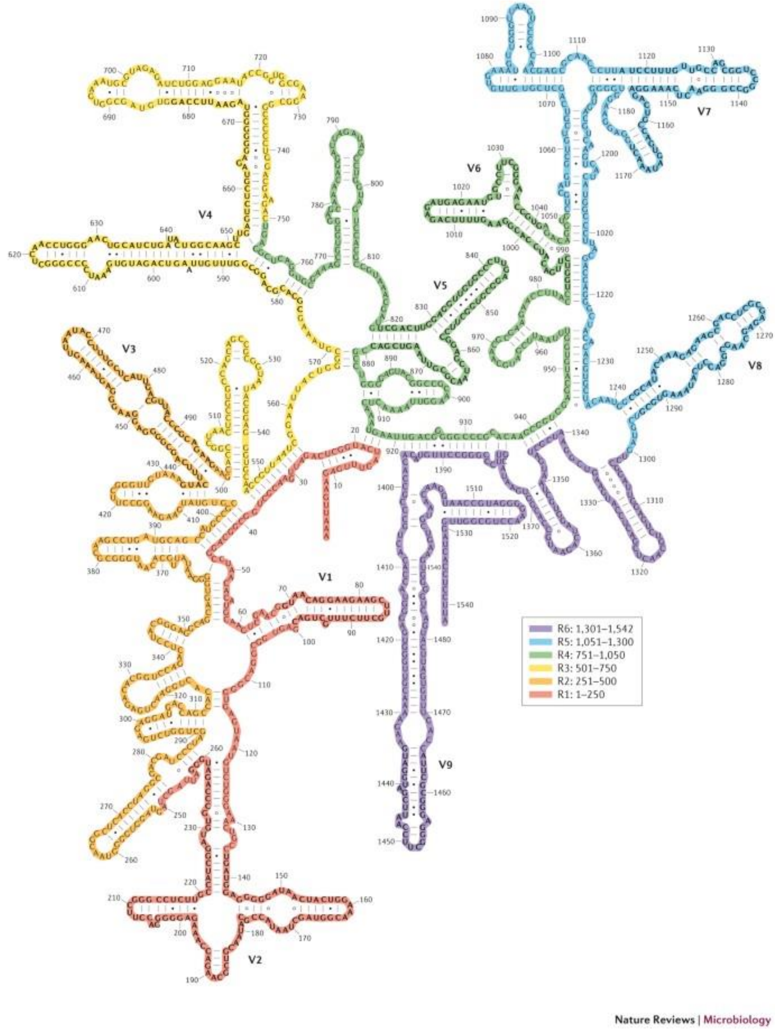


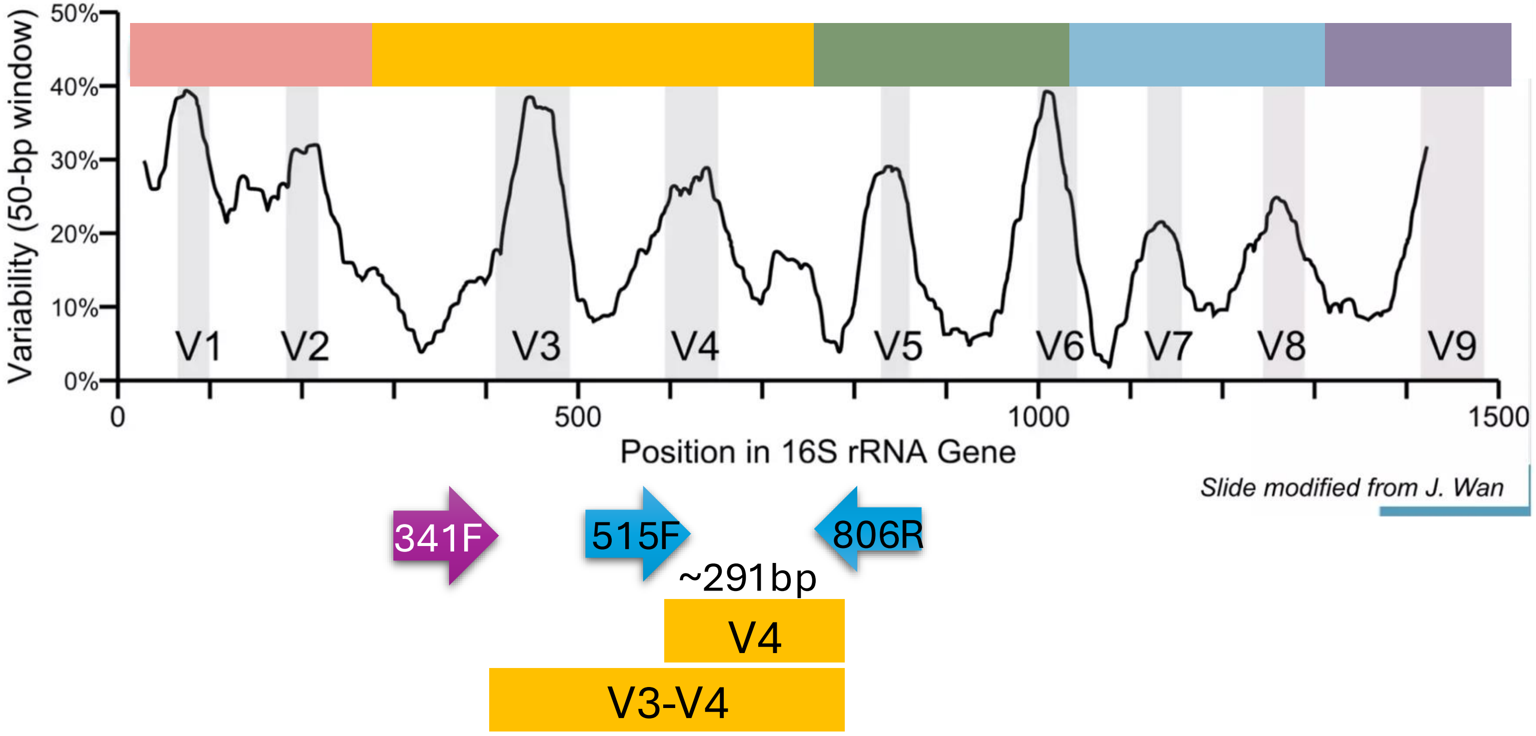


Sources:

https://doi.org/10.1073/pnas.87.12.4576
https://doi.org/10.1038/nmicrobiol.2016.48

16s rRNA structure





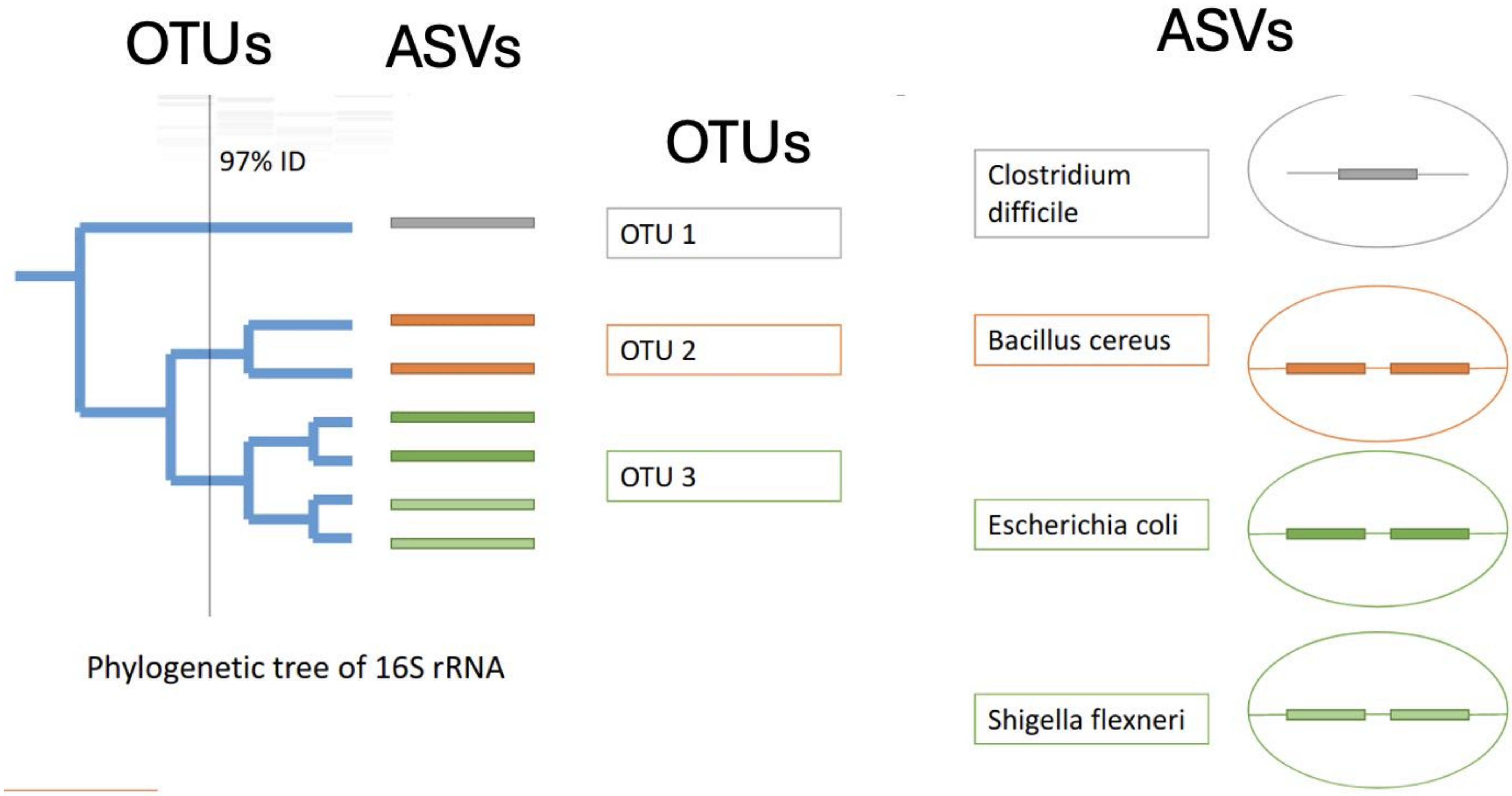
Source: https://doi.org/10.1038/nrmicro3330

Amplicon sequencing

OTUs	ASVs or ESVs
Operational **T**axonomic **U**nit	**A**mplicon/**E**xact **S**equence **V**ariant
The main differences are resolution, methodology, reproducibility.	
1 OTU ≥ **97%**	1 ASV = single nucleotide differences.

There has been a push since 2017 to only use ASVs (https://www.nature.com/articles/ismej2017119)

Amplicon sequencing



Source: Rue and Midoux, 2022

Amplicon Sequencing: Strengths & Limitations

When to use it

- Questions about diversity and/or compositional shifts.
- You need to process many samples cost-effectively (~$20* vs ~$150*)
- Low abundance organisms (PCR)

Bacteria and Archaea (16S rRNA)
Fungi (ITS, 18S rRNA)
Eukaryotes (18S rRNA)

When not to

- Species/Strain-level resolution or functional gene content.
- Gene copy number varies across taxa.

DNA/RNA virus
Archaea, bacteria, Eukaryotes

Software for Amplicon Analysis

QIIME 2

A plugin-based platform covering the whole workflow (Conda and terminal).

DADA2 (R package)

Directly in R for users who want full scripting control outside QIIME 2.

Mothur

One of the original open-source amplicon pipelines. **OTU**-based workflows.

- Quantitative insights into Microbial Ecology

- Pronounced like ”**chime”**.

- Created ~2010 during the Human Microbiome Project (2007 - 2016) under the leadership of Greg Caporaso and Rob Knight.

- Essentially, QIIME is a set of **commands** to transform microbiome **data** into **intermediate outputs** and **visualizations**.

QIIME 1 is outdated and QIIME 2 contains many improvements to analytic functionality, is easier to install, and is easier to use.

nature > nature methods > correspondence > article

Correspondence | Published: 11 April 2010

QIIME allows analysis of high-throughput community sequencing data

J Gregory Caporaso, Justin Kuczynski, Jesse Stombaugh, Kyle Bittinger, Frederic D Bushman, Elizabeth K Costello, Noah Fierer, Antonio Gonzalez Peña, Julia K Goodrich, Jeffrey I Gordon, Gavin A Huttley, Scott T Kelley, Dan Knights, Jeremy E Koenig, Ruth E Ley, Catherine A Lozupone, Daniel McDonald, Brian D Muegge, Meg Pirrung, Jens Reeder, Joel R Sevinsky, Peter J Turnbaugh, William A Walters, Jeremy Widmann, ... Rob Knight + Show authors

Nature Methods **7**, 335–336 (2010) | Cite this article

nature > nature biotechnology > correspondence > article

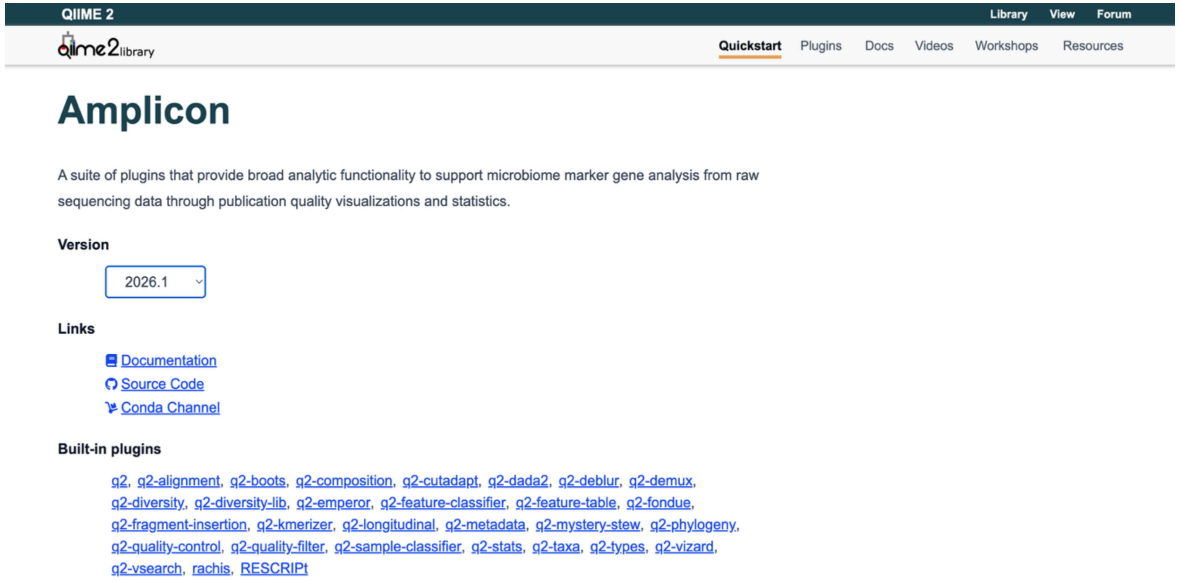
Correspondence | Published: 24 July 2019

Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2

Evan Bolyen, Jai Ram Rideout, Matthew R. Dillon, Nicholas A. Bokulich, Christian C. Abnet, Gabriel A. Al-Ghalith, Harriet Alexander, Eric J. Alm, Manimozhiyan Arumugam, Francesco Asnicar, Yang Bai, Jordan E. Bisanz, Kyle Bittinger, Asker Brejnrod, Colin J. Brislawn, C. Titus Brown, Benjamin J. Callahan, Andrés Mauricio Caraballo-Rodríguez, John Chase, Emily K. Cope, Ricardo Da Silva, Christian Diener, Pieter C. Dorrestein, Gavin M. Douglas, ... J. Gregory Caporaso + Show authors

Nature Biotechnology **37**, 852–857 (2019) | Cite this article

Qiime offers multiple tutorials and tools explanations


QIIME 2 Library View Forum

qiime2library Quickstart Plugins Docs Videos Workshops Resources

Amplicon

A suite of plugins that provide broad analytic functionality to support microbiome marker gene analysis from raw sequencing data through publication quality visualizations and statistics.

Version

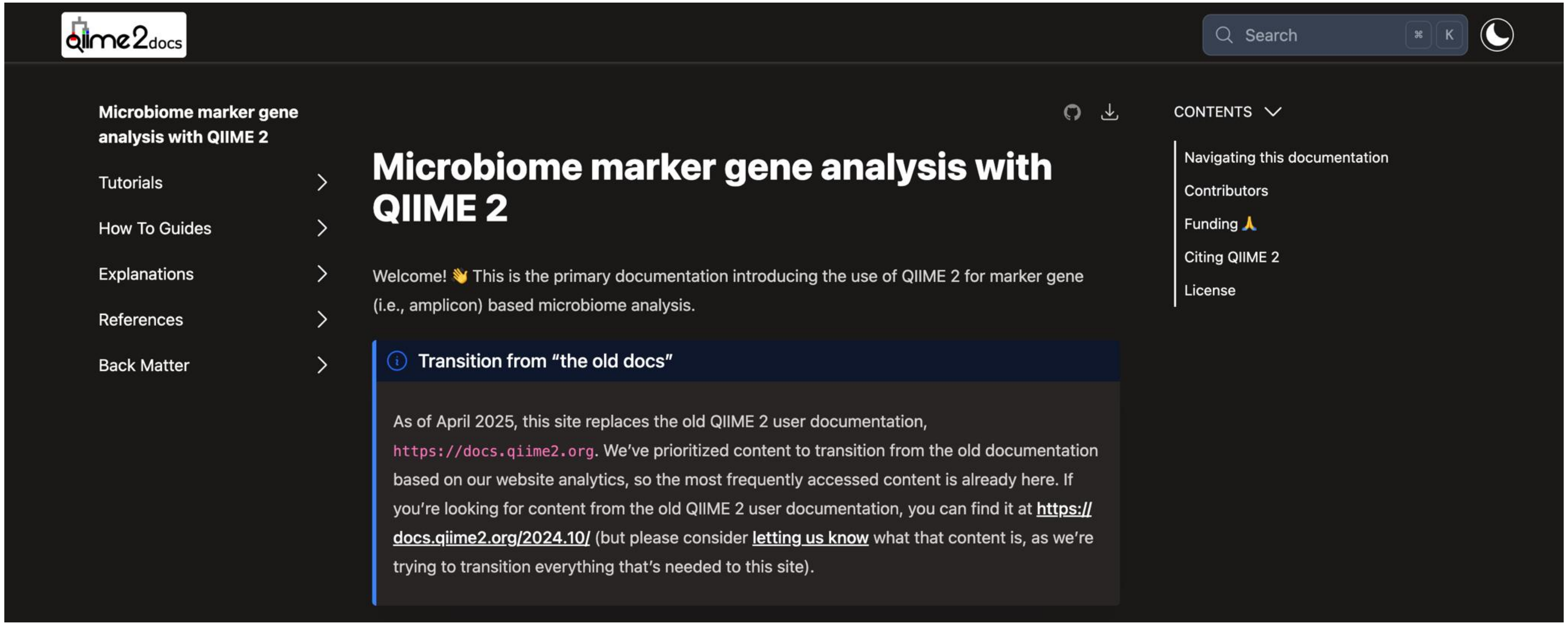
2026.1

Links

- Documentation
- Source Code
- Conda Channel

Built-in plugins

q2, q2-alignment, q2-boots, q2-composition, q2-cutadapt, q2-dada2, q2-deblur, q2-demux, q2-diversity, q2-diversity-lib, q2-emperor, q2-feature-classifier, q2-feature-table, q2-fondue, q2-fragment-insertion, q2-kmerizer, q2-longitudinal, q2-metadata, q2-mystery-stew, q2-phylogeny, q2-quality-control, q2-quality-filter, q2-sample-classifier, q2-stats, q2-taxa, q2-types, q2-vizard, q2-vsearch, rachis, RESCRIPt


qiime2docs Search ⌘ K

Microbiome marker gene analysis with QIIME 2

- Tutorials
- How To Guides
- Explanations
- References
- Back Matter

Microbiome marker gene analysis with QIIME 2

Welcome! 👋 This is the primary documentation introducing the use of QIIME 2 for marker gene (i.e., amplicon) based microbiome analysis.

Transition from "the old docs"

As of April 2025, this site replaces the old QIIME 2 user documentation, `https://docs.qiime2.org`. We've prioritized content to transition from the old documentation based on our website analytics, so the most frequently accessed content is already here. If you're looking for content from the old QIIME 2 user documentation, you can find it at https://docs.qiime2.org/2024.10/ (but please consider letting us know what that content is, as we're trying to transition everything that's needed to this site).

CONTENTS

- Navigating this documentation
- Contributors
- Funding 🙏
- Citing QIIME 2
- License

Qiime2 - What do you need

Using conda for installation.

“analogy”

conda installations

source code*

Sequencing reads for analysis

Type of reads

fastq/fq files → Raw reads. Files that contains nucleotide and their quality.

Single-end → <u>One</u> file per sample (sample.fq)

[Seq. technologies: Illumina, PacBio, Oxford Nanopore]

Paired-end → A <u>pair</u> of files per sample (sample_R1.fq, sample_R2.fq).

[Seq. technologies: Illumina]

Source

- Your own data
- Publicly available data from tutorials**
- The Sequence Read Archive stores raw reads from published studies.

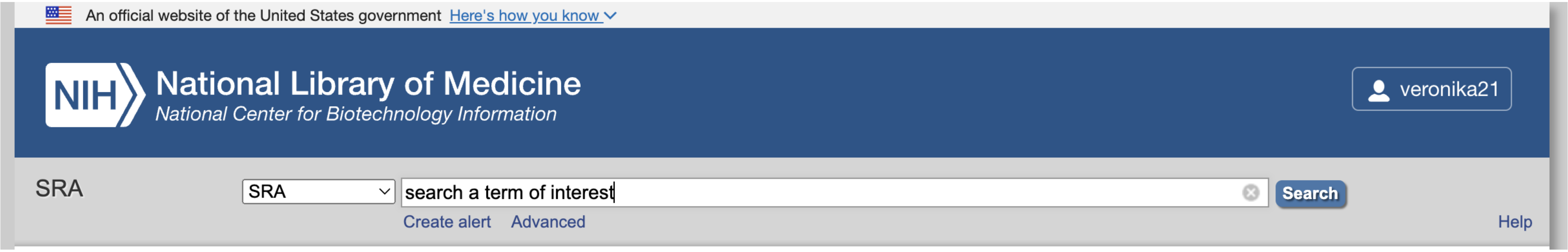
Databases

Once you have ASVs, you need a reference to assign taxonomy.

SILVA	16S/18S rRNA database spanning Bacteria, Archaea, and Eukaryotes. Updated in 2024 (https://www.arb-silva.de/).
GTDB	Genome Taxonomy Database, taxonomy standardized by genome phylogeny (https://gtdb.ecogenomic.org/).
***Greengenes2**	A 16S rRNA database re-built on a genome-based reference tree. Updated in 2024.
UNITE	The standard reference for fungal ITS sequences. (https://unite.ut.ee/).
NCBI	The largest general sequence archive. You can build a custom reference from it (https://www.ncbi.nlm.nih.gov/).

- QIIME 2 pre-trained classifiers for several of these are available at Qiime website.
- Check which version of Qiime you are using as that will affect what database you download.

The NCBI SRA (Sequence Read Archive)



What is SRA?

"The Sequence Read Archive (SRA) is the National Center for Biotechnology Information (NCBI) database that stores sequence data obtained from next generation sequence technology."

How is SRA useful?

Deposit your own data while submitting an article, Data mining (use data to ask different questions)

How to use SRA?

Online and command line (SRA tool kit: https://github.com/ncbi/sra-tools)

SRA

SRA | Search

Advanced

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Send to:

SRX22919344: MySeq of soil microbiome

1 ILLUMINA (Illumina MiSeq) run: 116,635 spots, 69.6M bases, 40.8Mb downloads

Design: The metagenomic DNA from each soil sample was extracted using the FastDNA spin Kit for soil from MP biomedical, following the manufacturers instruction. DNA concentration was determined by Nanodrop (Thermo Fisher) and the V3-V4 hypervariable 16s rRNA region (primers Pro341F 5-CCTACGGGNBGCASCAG -3 and Pro805R 5-GACTACNVGGGTATCTAATCC-3) was amplified. 16S rRNA library preparation was performed by BMR Genomics (Italy) for Illumina MiSeq 300PE sequencing.

Submitted by: University of Perugia

Study: Soil microbiota of organic and conventional management

PRJNA1053916 • SRP478556 • All experiments • All runs

show Abstract

Sample: Conventional low-input

SAMN38879569 • SRS19887352 • All experiments • All runs

Organism: soil metagenome

Library:

Name: 5A_Low_R2_G

Instrument: Illumina MiSeq

Strategy: AMPLICON

Source: METAGENOMIC

Selection: PCR

Layout: PAIRED

Runs: 1 run, 116,635 spots, 69.6M bases, 40.8Mb

Run	# of Spots	# of Bases	Size	Published
SRR27240947	116,635	69.6M	40.8Mb	2023-12-18

ID: 30984231

Recent activity

Turn Off Clear

soil metagenome (818729) SRA

metagenome (4013063) SRA

See more...

SRA | SRA | (soil microbiome) NOT amplicon[Strategy] | Search

Create alert Advanced Help

Full

SRX22885110: Soil metagenome

1 DNBSEQ (DNBSEQ-T7) run: 33.5M spots, 10G bases, 6Gb downloads

Design: normal metagenome of soil

Submitted by: Sichuan Normal University

Study: Soil metagenome Raw sequence reads

PRJNA1051770 • SRP477915 • All experiments • All runs

show Abstract

Sample: T3

SAMN38796427 • SRS19857384 • All experiments • All runs

Organism: soil metagenome

Library:

Name: S427

Instrument: DNBSEQ-T7

Strategy: WGS

Source: METAGENOMIC

Selection: RANDOM

Layout: PAIRED

Runs: 1 run, 33.5M spots, 10G bases, 6Gb

Run	# of Spots	# of Bases	Size	Published
SRR27205348	33,484,832	10G	6Gb	2023-12-14

ID: 30942389

Check point

- What is an amplicon and when to use it
- Why we use ribosomal sequences as markers
- What is OTU and ASV
- What software can we use
- Work with Qiime2

The QIIME 2 Pipeline

1	2	3	4	5	6
Import	**Denoise**	**Phylogeny**	**Diversity**	**Taxonomy**	**Diff. Abundance**
Raw FASTQ → .qza artifacts	error-corrects & merges reads → ASVs	Align ASVs & build a tree	Alpha & beta diversity metrics	Classify ASVs against a reference DB	ANCOM-BC2 flags significant taxa

The QIIME 2 Pipeline – Go to the terminal